

Grading and Feedback

Course: REI603M The AI Lifecycle

Assignment: Assignment 2 - Product Requirements Document

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Project: AI Math Coach for Handwritten Algebra

Grade: 10/10

No deductions. The PRD contains all required sections and the presentation slides were submitted on time.

Checklist

Required Sections

- [x] Executive Summary
- [x] Problem Statement
- [x] Target Users
- [x] Feature Requirements
- [x] Technical Architecture
- [x] Data Model
- [x] API Design
- [x] User Flow

Optional Sections (minimum 2 required)

- [x] Success Metrics
- [x] Risks and Mitigations

Other Requirements

- [x] Kano categorization of features (Must-Be, Performance, Attractive)
 - [x] User flow description
 - [x] Slides submitted in advance
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Strengths

The problem framing is precise: students need “confirmation, correction, or a small nudge” rather than full solutions. The explicit non-users section (students wanting quick answers, those without tablets) shows clear product positioning.

The answer leakage prevention design is thorough: structured prompts only, no free-text chat, post-processing to detect leaked answers, and CI tests for boundary violations. This addresses a core risk for any tutoring system.

The symbolic verification layer with explicit uncertainty handling (returning “uncertain” when confidence is low rather than guessing) demonstrates understanding of AI system limitations.

Teacher Feedback

1. **Answer leakage measurement:** How will you measure and monitor answer leakage in production? Define metrics and thresholds now. Consider logging all AI outputs and sampling for manual review.

2. **Agentic vision for labeling:** Google’s agentic vision capabilities in Gemini 3 Flash (<https://blog.google/innovation-and-ai/technology/developers-tools/agentic-vision-gemini-3-flash/>) can add visual labels directly to images. This could enable the AI to point at specific parts of a student’s work when giving feedback, improving clarity.
3. **Mistake analysis and reporting:** Your Mistake Library concept is good. Consider generating periodic reports showing mistake patterns over time. Parents would find “sign errors decreased 40% this month” compelling.
4. **Handwriting detection quality:** Good that you plan to detect and handle bad handwriting. The “rewrite step” prompt is the right fallback. Test extensively with real student handwriting, which varies more than you expect.
5. **Problem provision:** Will you offer problems to students, or only accept uploaded problems? Providing problems yourself would give you full context (you know the correct solution structure) and simplify the tutoring logic.
6. **Parent value proposition:** For B2C sales to parents, what is the compelling pitch? “Learning-focused tutoring” is abstract. Consider concrete claims: “students who use the coach improve test scores by X%” once you have data.
7. **Latency targets:** The 4-second latency target for hints is reasonable, but multimodal LLM calls can be slow. Consider background processing: query the LLM periodically as the student works, cache potential hints, then display cached responses when requested.
8. **Backend choices:** Python + FastAPI is solid. For easy scaling and auth, Firebase is worth considering. It handles authentication, real-time database, and hosting with minimal configuration.
9. **Model routing:** Consider using OpenRouter to swap between models easily. This lets you experiment with different models (OpenAI, Gemini, Anthropic) without code changes.
10. **Parent reporting:** Add a parent-facing view showing practice time and progress. Parents want visibility into whether their child is actually using the tool.
11. **Session handling:** Starting with session-based data (no persistent accounts) simplifies initial development. Build account persistence as a later feature once the core tutoring works.
12. **LaTeX conversion:** Converting handwritten math to LaTeX and showing the user for confirmation is smart. This creates a human-in-the-loop check that catches recognition errors before they propagate.
13. **Symbolic verification scope:** The PRD describes symbolic verification as “not a full CAS” but does not specify what operations it covers. Consider starting without symbolic verification and adding it only if LLM-only checking proves unreliable. You may not need it.

Suggestions for Implementation

- Build the Hint mode first, completely. This is your core interaction. Get it working reliably before adding Check Step or Reveal Solution.
- The mandatory cropping step is important for privacy but adds friction. Test whether students actually complete it correctly, or whether they crop poorly and send confusing images.
- Your success metrics are well-defined but ambitious. The “80% of hints lead to valid next step” target requires good hint quality and good step recognition. Both are hard.

- Consider a “confidence calibration” study: have the LLM rate its own confidence on step validation, then compare against ground truth. This helps you set thresholds for when to return “uncertain” vs a verdict.